

Software Requirement Specification (SRS)

Online Job Portal System

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Purpose

This document defines the requirements for the Smart Online Job Portal System—a comprehensive web platform designed to streamline recruitment.

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1. Introduction

1.1 Purpose

This document defines the requirements for the Smart Online Job Portal System—a comprehensive web platform designed to streamline recruitment. The platform will enable job seekers to search and apply for jobs while empowering employers to post vacancies and manage candidate applications efficiently. Additionally, the system includes administrative capabilities to manage users, moderate content, and generate real-time reports.

1.2 Scope

The portal will serve as a one-stop solution for:

- **Job Seekers:** Offering advanced search systems, personalized dashboards, application status tracking, and profile management.
- **Employers:** Facilitating job posting, applicant filtering, and direct communication with potential candidates.
- **Administrators:** Providing robust user management, system monitoring, and report generation tools to ensure overall system integrity and compliance.

1.3 Overview

The document details the functional and non-functional requirements, technical architecture, and user interface guidelines. It further outlines the risk management, change management processes, and future enhancement possibilities to adapt to evolving job market trends.

2. Overall Description

2.1 Product Perspective

The Smart Online Job Portal System is a stand-alone web application that will integrate with third-party services (e.g., resume parsers, email/SMS gateways) and support both desktop and mobile platforms through responsive design.

2.2 Product Functions

The key functionalities include:

- **Job Seeker Module:** Registration, profile management, job search system with advanced filters (from text or tags), application tracking, and status checking.
- **Employer Module:** Employer registration, job posting forum, application checker for reviewing candidate submissions, and detailed dashboards.
- **Job Posting Forum:** Should clearly mention the type of job with tags: (Internship/part time/blue collar/permanent or temporary job, freelance etc.) The forum must contain the requirements needed, deadline, location.
- **Admin Module:** User and role management, content moderation, monitoring system health, and generating performance and audit reports.

2.3 User Characteristics

- **Job Seekers:** Individuals looking for any type of job opportunities; require a simple, intuitive interface with robust search and tracking capabilities.
- **Employers:** HR professionals and hiring managers; need tools to post jobs and efficiently review applications.
- **Administrators:** IT staff responsible for overseeing platform operations, ensuring data integrity, and managing security protocols.

2.4 Constraints and Assumptions

- The system must be accessible 24/7 with high uptime.
- It must support high concurrent usage during peak job search periods.
- Data privacy and security will adhere to current regulatory standards.
- The platform will leverage modular design to allow future enhancements with minimal disruption.

3. System Modules and Functional Requirements

3.1 Job Seeker Module

3.1.1 Job Seeker Functionality

- **Registration & Profile Management:** Secure sign-up, login, profile editing, and resume upload.
- **Job Search System:**
 - Advanced search with filters (location, job type, experience, industry, etc.).
 - Real-time job listing updates.
- **Application Checker:**
 - Interface to track application status (e.g., applied, under review, shortlisted, rejected, or hired).
 - Historical log of all job applications.
- **Dashboard & Check Status:**
 - Personalized dashboard summarizing recent applications, notifications, and recommended jobs.
 - Quick view feature for application statuses with color-coded indicators.

3.2 Employer Module

3.2.1 Employer Details & Registration

- **Employer Registration & Profile Setup:** Secure registration, company profile, and verification procedures.
- **Job Posting Forum:**
 - Form-based job posting with fields for job description, qualifications, benefits, and deadlines.
 - Option to feature job posts for enhanced visibility.
- **Application Checker:**
 - Interface for reviewing incoming applications, filtering candidates, and sorting applications by status.
 - Bulk actions for scheduling interviews or sending automated responses.
- **Dashboard & Reporting:**
 - Interactive dashboard showing metrics (e.g., number of applicants, application trends, response time).
 - Reporting tools to export data and analyze hiring metrics.

3.3 Admin Module

3.3.1 Manage Users & System Monitoring

- **User Management:**
 - Role-based access control to manage job seekers, employers, and admin accounts.
 - Ability to activate, deactivate, or ban accounts, along with audit trail tracking.
- **Content Moderation:**
 - Tools to review and approve/reject job posts and applications to prevent spam and fraud.
- **System Monitoring:**
 - Real-time monitoring of system performance, usage statistics, and error logs.
 - Alerts and notifications for system anomalies or security breaches.
- **Report Generation:**
 - Comprehensive reports on user activity, job post performance, and overall platform usage.

4. Non-Functional Requirements

4.1 Performance Requirements

- The platform should support a high number of concurrent users with minimal latency. It should be designed in such a way that it can handle hundreds of users at a time.
- Job search and application processes should complete in under 3 seconds.

4.2 Security Requirements

- Secure authentication with encrypted credentials.
- Regular security audits and compliance with data protection regulations.
- Protection against common vulnerabilities (e.g., SQL injection, XSS).

4.3 User Registration & Authentication

- Users can register using email/phone.
- Verification via OTP or email confirmation.
- Admins can block/unblock suspicious users.

4.4 Usability and Accessibility

- Intuitive UI/UX with consistent design patterns and mobile responsiveness.
- Compliance with accessibility standards to support users with disabilities.
- Multi-language support, including at least English and one additional language.

4.5 Reliability and Availability

- 99.9% system uptime with robust backup and disaster recovery plans.
- Real-time data synchronization across modules.

4.6 Scalability and Extensibility

- Modular architecture to support additional features (e.g., AI-based job recommendations, video interviews).
- API-driven integration for third-party services.

5. System Architecture and Technical Requirements

5.1 Technology Stack

- **Frontend:** HTML5 and CSS 3 or React.js for dynamic user interfaces.
- **Backend:** Node.js and PHP for scalable API services.
- **Database:** MySQL for structured data.
- **Hosting:** Cloud-based services such as Google Cloud.
- **Integration:** RESTful APIs for integration with external resume parsers, email/SMS gateways, and social media platforms.

5.2 System Architecture Overview

- **Client-Server Model:** Web application accessible via browsers and mobile devices.
- **Microservices Architecture:** Each module (Job Seeker, Employer, Admin) can be developed, deployed, and scaled independently.
- **Data Flow:** Secure data transmission with SSL/TLS; centralized logging and analytics.

6. Quality Assurance and Testing

6.1 Testing Strategy

- **Unit Testing:** Ensure each module functions as expected.
- **Integration Testing:** Verify communication between modules (e.g., job posting and application checking).
- **System Testing:** Full end-to-end testing in real-world scenarios.
- **Security Testing:** Regular vulnerability assessments and penetration testing.

6.2 User Acceptance Testing (UAT)

To engage users to validate the system usability and performance.

7. Maintenance and Monitoring

7.1 Monitoring Strategy

- **Real-Time System Monitoring:** We have to track server performance, user activity, and error rates.
- **Audit Logs:** Detailed logging for user actions, data changes, and administrative tasks.
- **Incident Management:** Predefined protocols for addressing downtime, security incidents, or performance issues.

7.2 Maintenance

- Regular updates and patches based on system analytics.
- Scheduled system health checks.

8. Risk Management and Change Management

8.1 Risk Management

- **Identified Risks:** Data breaches, downtime during peak usage, fraudulent job posts.
- **Mitigation Strategies:** Regular security audits, load testing, and continuous monitoring.
- **Contingency Plans:** Backup systems, emergency support teams, and rapid incident response protocols.

8.2 Change Management

- **Change Request Process:** Formal logging of requests, impact analysis, and approval workflows.
- **Documentation:** Updated SRS and technical documents reflecting changes.
- **Stakeholder Communication:** Regular meetings and updates to ensure transparency.

9. Future Enhancements

- **AI-Based Job Recommendations:** Leveraging machine learning to match job seekers with suitable positions.
- **Video Interview Integration:** Embedding real-time video interview capabilities within the platform.
- **Advanced Analytics:** More sophisticated data visualization and predictive analytics for hiring trends.

10. Appendices

10.1 Glossary of Terms

A list of technical and domain-specific terms used throughout this SRS.

- **SRS:** Software Requirement Specification
- **UI:** User Interface
- **UX:** User Experience
- **API:** Application Programming Interface
- **DBMS:** Database Management System
- **HR:** Human Resource
- **IT:** Information Technology
- **SQL:** Structured Query Language
- **UAT:** User Acceptance Testing
- **SMS:** Short Message Service
- **XSS:** Cross Site Scripting
- **AI:** Artificial Intelligence
- **SSL:** Secure Sockets Layer
- **TLS:** Transport Layer Security
- **PHP:** Hypertext Pre-Processor
- **HTML 5:** Hypertext Markup Language
- **CSS 3:** Cascading Style Sheet

10.2 Issues

API integration challenges for third-party tools, handling more users than expected can break the system.

10.3 Reference Documents

- Previous SRS documents for similar systems.
- Industry standards.